Siddharth Sriram Sikha

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EDUCATION

Texas A&M University - MS Mechanical Engineering, (Minor: Project Management) – GPA: 3.66/4.0 **Aug 2023 – May 2026**

Vellore Institute of Technology - Bachelor of Technology in Mechanical Engineering – GPA: 3.9/4.0 **Aug 2019 – May 2023**

SKILLS

Core Design Tools: SolidWorks (CSWA), Siemens NX, Creo, Fusion 360, Revit (Basic), AutoCAD, Technical drawings, GD&T (ASME Y14.5)
Simulation & Analysis: ANSYS Workbench (Static, Thermal), Abaqus, MATLAB, FMEA, SPC, RCA, DFM, DFA, HVAC & Industrial systems
Manufacturing Processes: Additive, subtractive, forming and joining, Machining & fabrication fundamentals, Tolerance stackup analysis
Project Management: Scope definition, cost estimation using BOM's, schedule tracking, Risk Register Development, Monte Carlo Simulation (@Risk), NPV & IRR Financial Modeling, PDRI Assessment, RSMeans Cost Estimation, Schedule Risk Analysis
Programming & Documentation: Python (Data analysis, Automation), Mathcad, LaTeX, MS Office, Teamcenter, Windchill PLM, SAP

WORK EXPERIENCE

Texas A&M University **Aug 2024 – Present**
Graduate Teaching Assistant *College Station, United States*

- Mentored 120+ students across *Product Design & Solid Modeling* and *Metallic Materials* courses, teaching Fusion 360 and guiding hands-on labs in hardness testing, heat treatment bridging CAD design with practical materials engineering

SLB(Schlumberger) **May2024 – August 2024**
Mechanical Engineering Intern, Completions Dept *Houston, United States*

Project1 – Load Cell Integration for Packer Testing

- Increased force measurement accuracy by 7% by designing and integrating a custom load cell into a 150 ksi hydraulic ram system, replacing a pressure transducer and validating load paths through stress analysis and 3D modeling in CREO
- Improved manufacturability and part alignment by leading cross-functional design reviews and executing FMEAs, resolving tolerance conflicts and minimizing required assembly modifications

Project 2 – FEA-Based Optimization of Hydraulic Packer

- Improved component fatigue life and reduced peak stresses by 12% through FEA-driven geometry optimization in ANSYS, cutting manufacturing defects by 11% and material costs by 4.5% using standardized workflows in GEMS and Windchill PDM

Tecumseh Products **May 2022 – July 2022**
Mechanical Design Intern *Hyderabad, India*

- Conducted load calculations, performance modeling, and energy efficiency assessments for condenser and refrigeration systems, applying thermodynamics and heat transfer principles
- Designed and optimized HVAC subsystems including condensers, refrigerant loops, and airflow components; created 3D CAD assemblies and 2D layouts using Siemens NX and basic exposure in Revit for ductwork, piping, and mechanical room layouts
- Collaborated with cross-functional teams (manufacturing, electrical, suppliers) to ensure seamless integration of mechanical systems into product and facility requirements, aligning designs with clients adhering to ASHRAE standards in the United States

PROJECTS

Structural Redesign of Playground Equipment Under Dynamic-Load Conditions (*Skills: FEA, ANSYS, SolidWorks, DFM)*

- Redesigned playground structures for combined adult + child dynamic loads, validating static stress, deflection, and FOS > 2.7 via SolidWorks Simulation, ANSYS Workbench, and hand-calc/FEA correlation within 5–15 %
- Applied DFM to remove excess material while preserving integrity, delivering lighter, safer, and more cost-effective components

Evaluation of Vibration Techniques to Measure Scuffing in Packaged Products *(Skills: SolidWorks Design, 3D printing, Data analysis)*

- Designed and prototyped a scuffing rig using SolidWorks and 3D printing (Ultimaker 5S, Cura slicing) with bushings and springs to simulate real-world vibration effects on packaged products, addressing friction and alignment issues through iterative design improvements
- Conducted vibration testing using a Lansmont Saver9X30 data logger and packaging vibration table, applying (PSD) analysis to compare field-collected data with lab-based simulations

EPC Project Feasibility and Risk Analysis – *(Skills: @Risk Simulation, RSMeans Estimation, Financial Modeling, PDRI Framework)*

- Performed Front-End Planning (FEP-1) for a $30 M two-building real estate investment, integrating business objectives, cost, and risk factors to enable board-level Go/Hold/Kill decisions
- Developed RSMeans-based cost models (±25–40 % accuracy) and executed Monte Carlo risk simulations (50 k iterations) in @Risk to establish P50/P80 confidence and support capital planning
- Conducted financial and project readiness analysis using IRR, NPV, DSCR, and PDRI (0.655) metrics, identifying Phoenix, AZ as the optimal site meeting the ≥ 10 % MARR target and qualifying for FEP-2

ACHIEVEMENTS

- Cleared the FE Mechanical Exam (NCEES); pursuing EIT certification upon graduation
- Fully funded Teaching Assistantship under Prof Chris Smith & David Craig from the ETID department – Texas A&M